

Module - 3 : Corrosion & Electrode System

Corrosion:

"The gradual degradation @ deterioration of metal, which form its stable oxide form, whenever it is exposed to corrosive environment (oxygen & moisture) is called as corrosion.

Ex: Rusting of Iron, green deposit of CuO on Cu etc.

The ill effects of corrosion are listed as follows,

- * Loss of efficiency
- * Damage of metallic equipments.
- * Inability to use of metallic materials.
- * Loss of valuable materials such as blockage of pipes, mechanical damage of underground water pipes.
- * causes pollution due to escaping products from corrosion
- * Depletion of natural resource (Metals).

The global cost of corrosion is estimated to USD 2505 million, which is equivalent to 3.4% of the global GDP.

Corrosion mainly classified into two types.

Dry corrosion: corrosion occurs due to the attack of dry gases without moisture/aqua media..

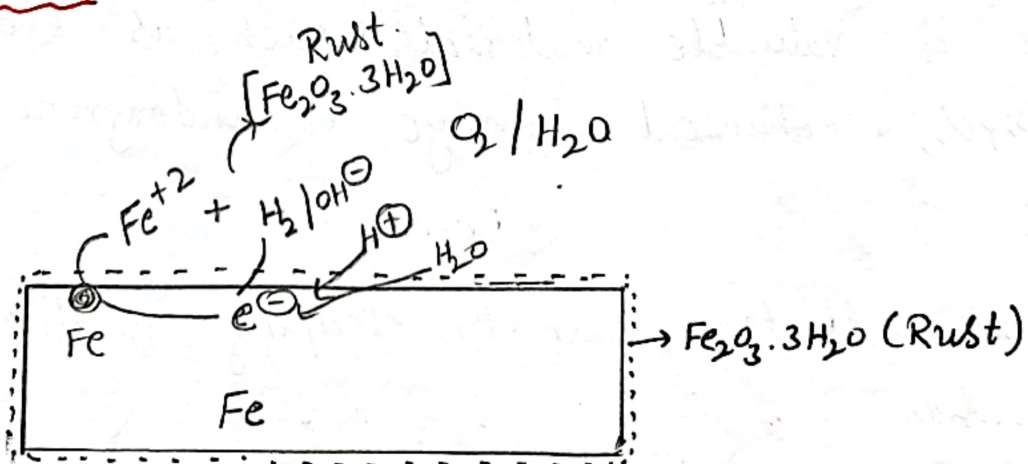
Ex: Dry chlorine (Cl_2) gas attack on metal causes corrosion

Dry corrosion can be explained on the basis of chemical reactions.

Wet corrosion: corrosion occurs due to the presence of moisture @ aqueous media Ex: Rusting of Iron.

Wet corrosion can be explained on the basis of electrochemical theory.

Electrochemical Theory of corrosion



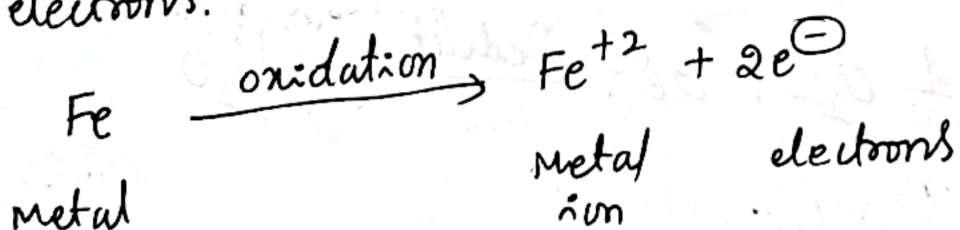
When Iron metal exposed to corrosive environment such as oxygen and moisture, large number of minute galvanic cells are produced over the metal surfaces.

Each galvanic cell differentiated as two imaginary layers such as anodic & cathodic layers.

According to the electrochemical theory, anodic layer undergoes oxidation, which forms metal ions from metal with the liberation of electrons. The liberated electrons from anode migrate towards cathodic area, it undergoes reduction.

At this time, due to the migration of electrons from anodic area to cathodic area, which develops a potential difference between them. The developed potential difference is a driving force for corrosion process. More potential difference occurs, which increases the corrosion of metal.

At Anode: At anode, iron metal undergoes oxidation, which forms its metal ions with the liberation of electrons.



At cathode: The liberated electrons from an anode are received by the cathode, which undergoes reduction. Hydrogen is liberated at the cathode.

The cathodic reactions are takesplace in 2 ways,

① Liberation of Hydrogen

② Absorption of oxygen

① Liberation of Hydrogen (in the absence of O_2)

(a) Acidic media

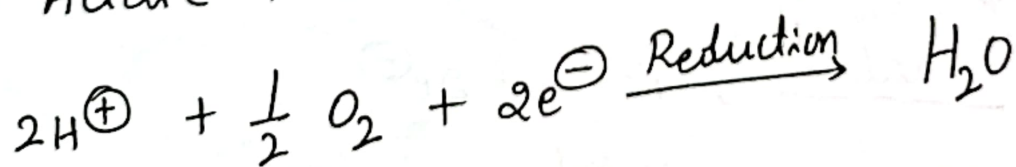


(b) Neutral Media

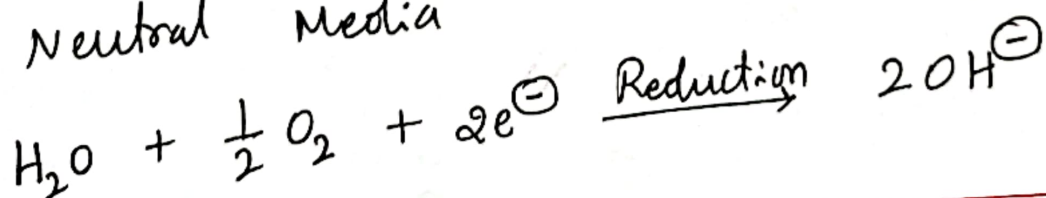


② Absorption of oxygen (in the presence of O_2)

(a) Acidic media

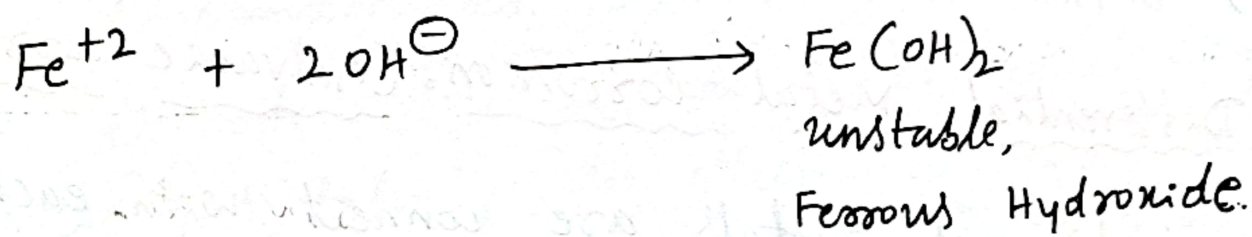


(b) Neutral media

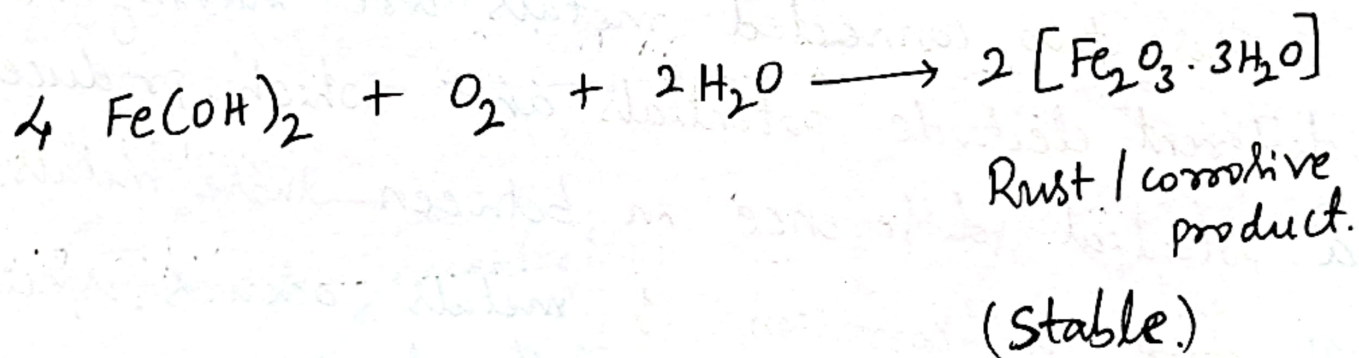


Hydrogen (H_2) and Hydroxyl ions ($OH^\ominus$) liberated at the cathode are called as cathodic ions.

Anodic ion (Fe^{+2}) combines with any one of the cathodic ion ($H_2/OH^\ominus$) to form unstable Ferrous hydroxide [$Fe(OH)_2$].



unstable ferrous hydroxide further undergoes oxidation in the presence of oxygen and moisture to form a stable ~~ferric~~^{Tri} hydrated ferric oxide [$Fe_2O_3 \cdot 3H_2O$] as a rust. @ corrosive product



Types of corrosion

Corrosion can be classified into various types, among those, are

- 1) Differential Metal corrosion.
- 2) Differential Aeration corrosion.
- 1) Differential Metal corrosion: (Galvanic corrosion)

Two different metals are connected with each other and exposed to a corrosive environment, the metal which leads to the corrosion is called as differential metal corrosion.

These two connected metals are having different electrode potentials and which produces a potential difference in between those metals. Therefore, the corrosion of metals occurs, especially anodic metal undergoes corrosion.

The couple of two metals involved in this type of corrosion is called as galvanic couple, hence this corrosion is also called as galvanic corrosion.

When two different metals (different potentials) like zinc and copper are electrically connected and are exposed to a corrosive environment such as oxygen and moisture, anodic metal undergoes corrosion. In this couple of metals Zn has lesser electrode potential and higher reactivity, which is easily undergoes corrosion. And another metal Cu is having more electrode potential, and lesser reactivity, which is protected by the attack of corrosion. A potential difference developed in between Zn & Cu, proceeds corrosion of anodic metal such as Zn. ($\therefore$ potential difference is a driving force for corrosion process).

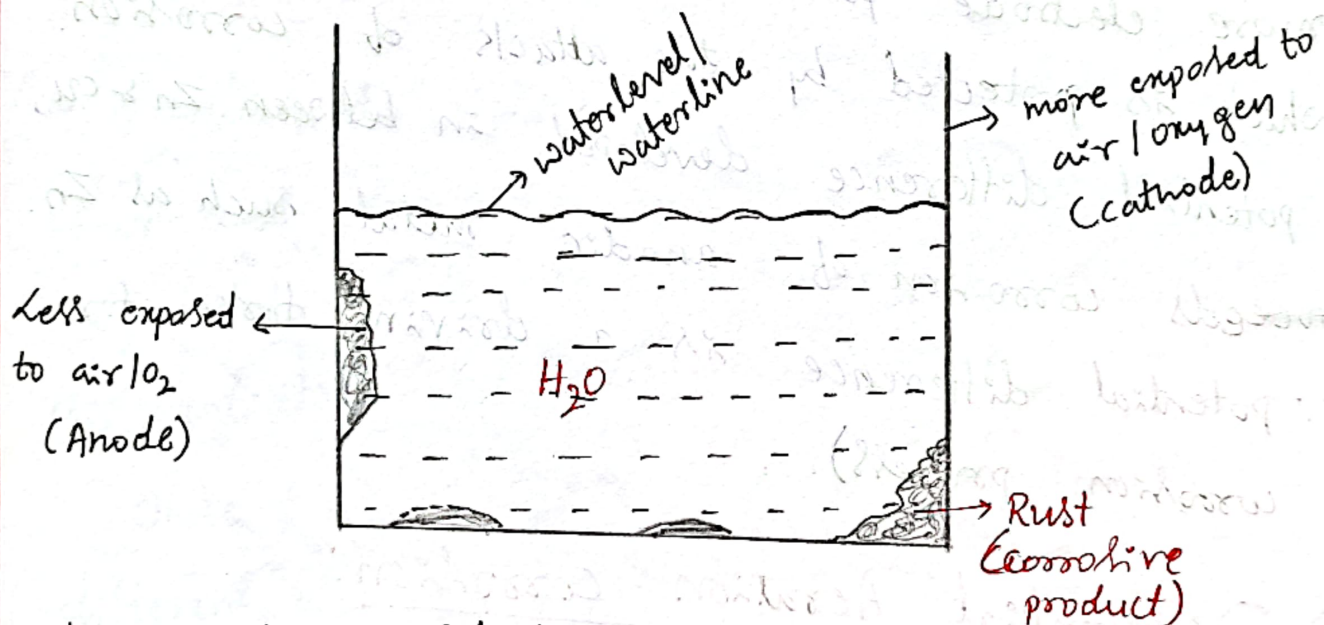
2) Differential Aeration corrosion:

The different concentration of Air/oxygen exposed to two different surface areas of a same metal, which proceeds the corrosion is called as Differential Aeration corrosion.

One part of the metal surface is in contact with lesser concentration of air/oxygen, it acts as an anodic area, which is easily undergoes corrosion.

(corrosion is nothing but oxidation of metal, oxidation occurs at anode). The another surface part of the same metal is exposed to higher concentration of air @ oxygen which acts as a cathode and it is unaffected by the attacks of corrosion.

The differential aeration corrosion can be explained with waterline corrosion as follows.



when water is filled into a metal tank (Fe), which is exposed to corrosive environment such as oxygen & moisture. The surface area ~~above the~~ below the water level is in contact with ~~higher~~ lesser concentration of air/oxygen, it acts as an anode, which is ~~at~~ easily undergoes corrosion.

The surface area above the water level, which is exposed with higher concentration of air/oxygen.

it acts as a cathode, ^{and which} is unaffected by the attack of corrosion. Therefore anodic area is called as corroded area and cathodic area is called as protected area by corrosion. A potential difference developed in between anode and cathode, which proceeds the corrosion of anodic area.

Corrosion Control

Corrosion of metals controlled by various methods by avoiding the formation of galvanic cell over the metal surface, when it is in contact with corrosive environment. Some of the corrosion controlling methods are

* Galvanisation.

* Anodisation.

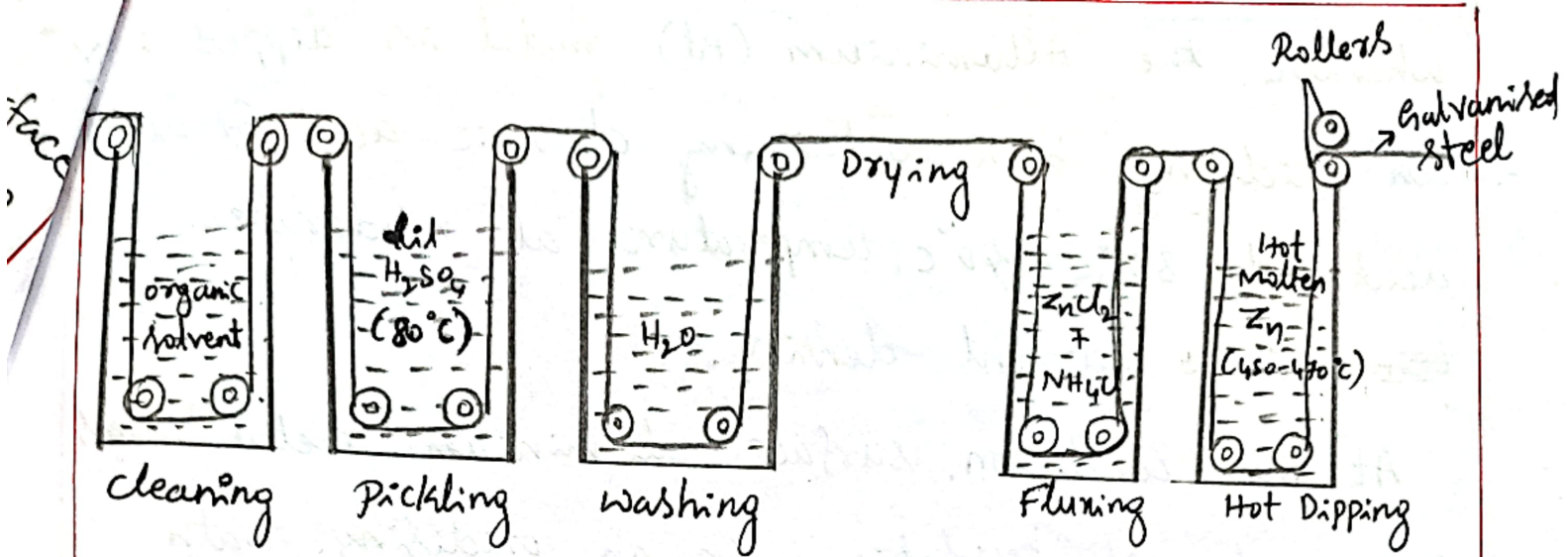
* Sacrificial Anodic method.

Dr. B. M. PRASANNA
Associate Professor
Department of Chemistry
Bapuji Institute of Engineering and Technology
DAANAGERE - 577 004.

Galvanisation

Thin film of Zn metal is coated onto the surface of base metal to be protected by the attacks of corrosion is called as galvanisation. The galvanisation process involves following steps,

- * The base metal is cleaned by using organic solvents such as acetone, alcohol etc.-.
- * cleaned base metal is treated with dil. H_2SO_4 at $80^\circ C$ for about 10 minutes to removal of initial rust present onto the metal surfaces. This process is called as pickling process.
- * Base metal is washed with distilled water and dried at room temperature.
- * clean and dry base metal is treated with flux materials such as $ZnCl_2$ & NH_4Cl .
- * The ~~base metal~~ base metal is dipped into hot molten Zn at $450 - 470^\circ C$ temperature, a thin layer Zn is coated over the steel surfaces. This method is called as hot dipping method.
- * The excess Zn can be removed by rolling or wiping technique.

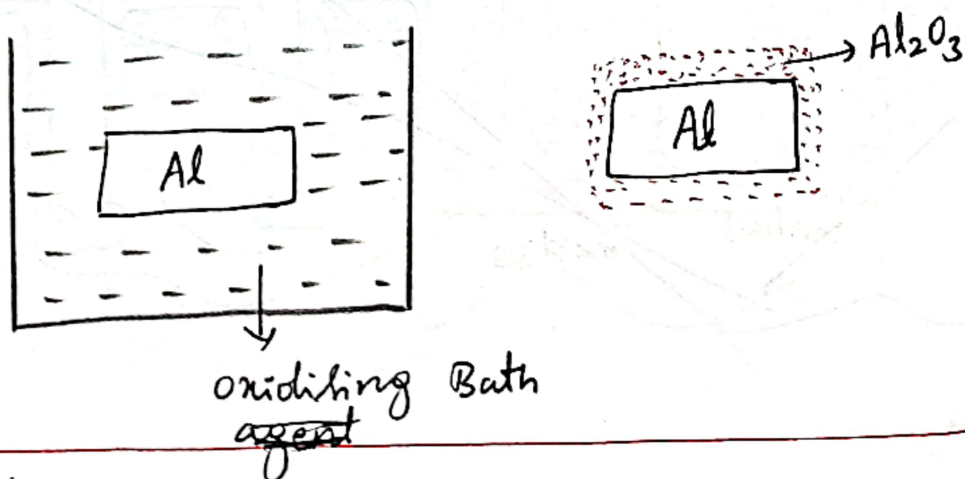


Anodisation

The phenomenon of formation of a thin, passive film of its respective metal oxide formed over the metal surface as a protective coating is called as anodisation.

Anodisation method is use to protect highly reactive metals such as aluminium, Zinc & magnesium etc-

Anodisation of Aluminium:



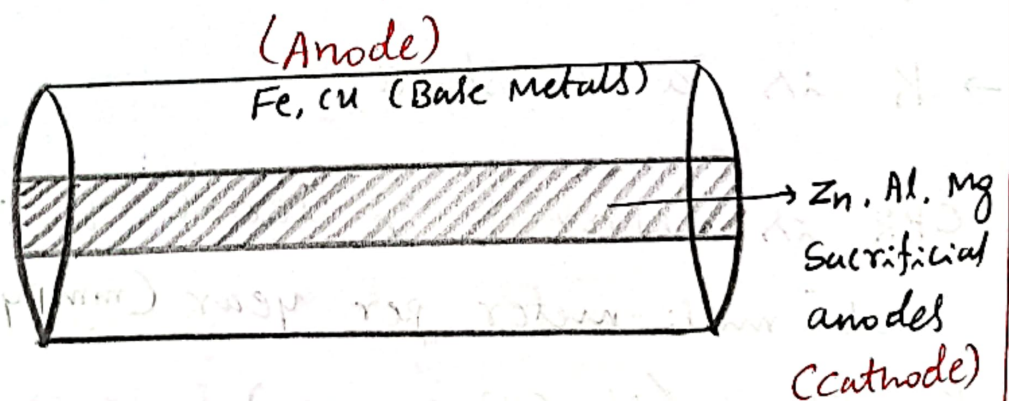
whenever the Aluminium (Al) metal is dipped in an oxidising bath containing chromic acid or oxalic acid at $35^{\circ}\text{C} - 40^{\circ}\text{C}$ temperature at moderate ~~temperature~~ current density.

At this condition, surface aluminium metal atoms undergo oxidation in an oxidising bath to form a thin passive film of aluminium oxide (Al_2O_3). This produced layer of aluminium oxide acts as a protective layer, which cuts off the interaction between the base metal and corrosive environment, it protects the metal by attack of corrosion. Anodised metal articles are used as window frames, curtain carriers and soap boxes etc.-.

Sacrificial Anodic Method

The process of protecting a metal by the attack of corrosion by making the entire metal converted as cathode is called as cathodic protection.

Sacrificial anodic method is a type of cathodic protection. In this method a metal is to be protected [Ex: Iron] is connected to a more reactive metal like Zn, Mg, Al etc. Here highly reactive metals acts as an anode and easily undergoes corrosion. The actual base metal is converted as cathode, and it is unaffected by the attack of corrosion. The highly reactive metals destroy itself to protect the metals are called as sacrificial anodes.



Corrosion Penetration Rate (CPR)

Corrosion Penetration Rate (CPR) is a speed at which any metal deteriorates in specific corrosive environment. Corrosion penetration rate depends upon metals and environmental conditions. And also corrosion Penetration Rate (CPR) is the amount of ~~corrosion~~^{metal} loss in thickness per unit of time. CPR can be calculated by the following expression,

$$\text{CPR} = \frac{K W}{P A t}$$

- Where 'W' is the weight loss after exposure time "t"
- P & A represents Density and exposed area of specimen respectively.
- K is a constant.

CPR is expressed in terms of milli per year (mpy) and milli meter per year (mmpy). Therefore,

$$K = 534 \quad (\text{If CPR in mpy})$$

$$K = 87.6 \quad (\text{If CPR in mmpy})$$

		CPR (mpy)	CPR (mmpy)	
Constant	K	534	87.6	
Weight loss	W	mg	mg	
Density	ρ	g/cm^3	g/cm^3	
exposed Area of specimen	A	inch^2	cm^2	
Time	t	hours	hours	

Numerical problems on CPR:

1) A thick brass sheet of area 400 inch^2 is exposed to air near the ocean. After 1 year period it was found to be experience a weight loss of 375 gms due to the corrosion. If the density of brass is 8.73 g/cm^3 . calculate the corrosion penetration Rate in mpy and mmpy.

Solution:

	(mpy)	(mmpy)
K	534	534
W	$375 \times 1000 \text{ mg}$	$375 \times 1000 \text{ mg}$
ρ	8.73 g/cm^3	8.73 g/cm^3
A	400 inch^2	$400 \times (2.54)^2 \text{ cm}^2$
t	$-(1\text{-year}) \quad 24 \times 365 \text{ hours}$	$24 \times 365 \text{ hours}$

$$CPR = \frac{KW}{PA \cdot t} = \frac{534 \times 375 \times 1000}{8.73 \times 400 \times 24 \times 365} = 6.546 \text{ mpy} \quad \text{Calc}$$

$$CPR = \frac{KW}{PA \cdot t} = \frac{87.6 \times 375 \times 1000}{8.73 \times 400 \times (2.54)^2 \times 24 \times 365} = 0.166 \text{ mmpy}$$

Q) Calculate the CPR in both mpy and mm/y for a thick steel of area 100 inch^2 which experiences a weight loss of 485 gms after one year. (Density of steel = 7.9 g/cm^3).

~~Given~~ Given. CPR in mpy CPR in mm/y

k	534	87.6
W	485 gms $485 \times 1000 \text{ mg}$	$485 \times 1000 \text{ mg}$
p	7.9 g/cm^3 7.9 g/cm^3	7.9 g/cm^3
A	100 inch^2 100 inch^2	$100 \times (2.54)^2 \text{ cm}^2$
t	1 year $365 \times 24 \text{ hrs}$	$365 \times 24 \text{ hrs}$

$$\text{CPR} = \frac{kW}{pAt} = \frac{534 \times 485 \times 1000}{7.9 \times 100 \times 365 \times 24} = 37.42 \text{ mpy}$$

$$\text{CPR} = \frac{kW}{pAt} = \frac{87.6 \times 485 \times 1000}{7.9 \times 100 \times (2.54)^2 \times 365 \times 24} = 0.951 \text{ mm/y}$$

$$\text{CPR} = \frac{W}{pA \cdot t} = \frac{485 \times 1000}{7.9 \times 100 \times 365 \times 24} = 37.42 \text{ mpy}$$

$$\text{CPR} = \frac{W}{pA \cdot t} = \frac{485 \times 1000}{7.9 \times 100 \times (2.54)^2 \times 365 \times 24} = 0.951 \text{ mm/y}$$

	CPR (mpy)	CPR (mmpy)
K	537	87.6
W	mg	mg
P	g/cc	g/cc
A	inch ²	cm ²
t	hrs	hrs

3) A Thick brass sheet of area 400 inch² is exposed to air near the ocean. After 1 year period it was found to be experience a weight loss of 375 gms due to corrosion. If the density of brass is 8.73 g/cm³ calculated the corrosion penetration rate in mpy and mmpy.

	given.	mpy	mmpy
K			
W	375 gms	375 x 1000 mg	375 x 1000 mg
P	8.73 g/cm ³	8.73 g/cm ³	8.73 g/cm ³
A	400 inch ²	400 inch ²	400 x (2.54) ² cm ²
t	1 year.	24 x 365 hrs	24 x 365 hrs

$$CPR = \frac{KW}{PA t} = \frac{537 \times 375 \times 1000}{8.73 \times 400 \times 24 \times 365} = 6.546 \text{ mpy}$$

$$CPR = \frac{KW}{PA t} = \frac{87.6 \times 375 \times 1000}{8.73 \times 400 \times (2.54)^2 \times 24 \times 365} = 0.166 \text{ mmpy}$$

Conductometry

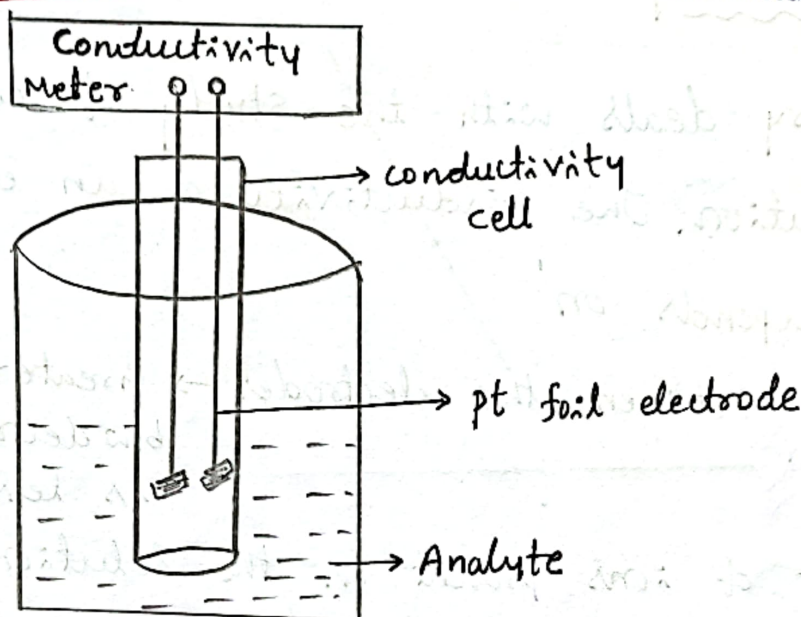
Theory:

Conductometry deals with the study of conductance in a solution. The conductivity of an electrolyte solution depends on

- * Distance between the electrodes $\rightarrow$ Greater distance b/w electrodes, conductivity is less
- * Number of ions present in the solution $\rightarrow$ Higher the concentration of the solution higher is the conductivity
- * The mobility of the ions $\rightarrow$ Smaller the ion, mobility and conductivity is greater.
- * The temperature of the solution $\rightarrow$ Higher the temperature, increase the mobility and conductivity

The conductivity of an electrolytic solution depends measured by using a conductometer. The conductance of a solution is directly proportional to the number of ions @ concentration of the solution.

Instrumentation:



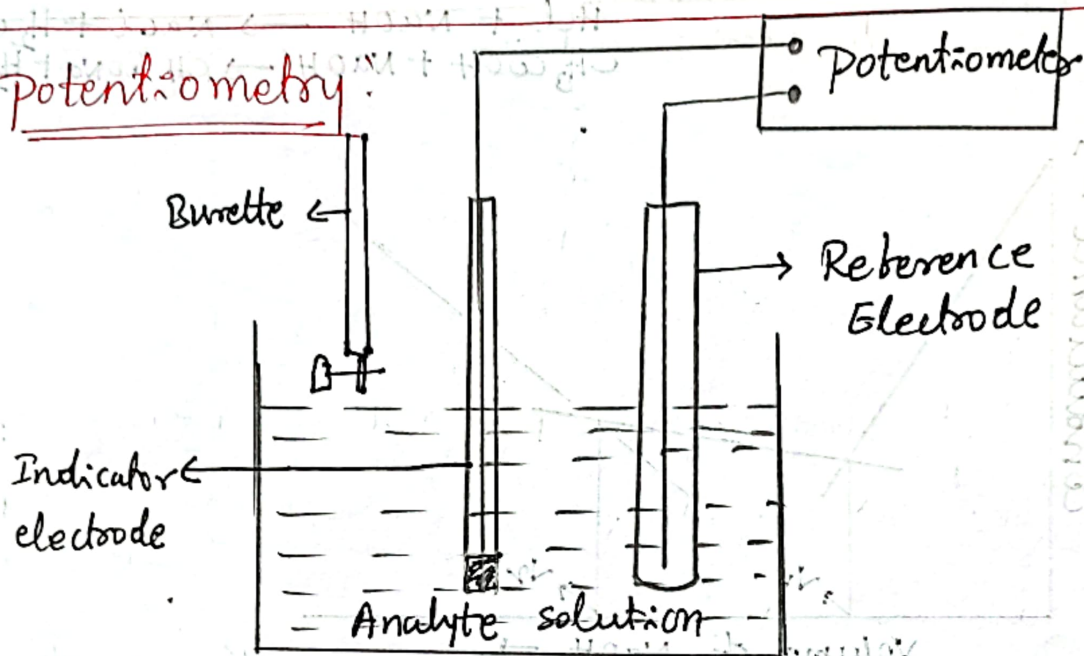
Two platinum foils placed apart at 1 cm, is immersed in the test solution at constant temperature, and which is connected to a conductivity meter. The conductivity varies by the addition of titrant.

Applications:

- * colored solutions can be titrated
- * It has high accuracy than old methods.
- * It can be used for weak acids & weak bases.

BM:

Potentiometry



Theory

When the pair of electrodes [Reference and Indicator electrode] are immersed in the solution, which shows the potential difference by the addition of titrant by the change in the concentration of the ions.

The potential developed in between the electrodes depends upon the concentration of the electrolyte, which is given by the Nernst's equation.

$$E = E^{\circ} + \frac{RT}{nF} \ln [M^{+n}]$$

For redox reaction, the above equation becomes,

$$E = E^{\circ} + \frac{RT}{nF} \ln \frac{[Fe^{+3}]}{[Fe^{+2}]}$$

The equivalence point can be determined by using characteristic graph.

Instrumentation: The basic components of potentiometer

- are,
- 1) An indicator electrode
 - 2) A reference electrode
 - 3) ~~Salt bridge~~

A typical electrochemical cell for potentiometric studies is constructed by coupling two electrodes. one electrode is a reference electrode (Ex: calomel electrode) and

another one is indicator electrode (Ex: platinum electrode).

Both electrodes are immersed in a given electrolyte to measure potential of the solution with the addition of titrant.

Potentiometric Estimation of Iron

When a solution of Ferrous ammonium sulfate was titrated against a strong oxidising agent like $K_2Cr_2O_7$ solution. The $K_2Cr_2O_7$ oxidises iron from ferrous state (Fe^{+2}) to Ferric state with the liberation of electrons. The liberated electrons were migrated from anodic to cathodic area, it develops a potential difference $[E]$. By using potential value, change in potential $[\Delta E]$ can be measured. Finally plot a graph of $\Delta E/\Delta V$ against $\Delta E/\Delta V$, to get an equivalence point.

Normality of an FAS can be measured by equivalence point & it will be multiplied by an equivalent weight of an FAS (392) to get amount of iron present in the given solution.

